A Ring-Like Excess of Galaxies at an Angular Radius of 25.5° around the CMB Dipole Direction: Data from 2MRS and Cosmicflows-4

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Abstract

We report the detection of a ring-like excess in the surface density of galaxies at an angular radius of 25.5° around the direction of the cosmic microwave background (CMB) dipole, with coordinates RA = 167.9°, Dec = −7.2°. The analysis is performed independently in two catalogs, 2MRS and Cosmicflows-4 (CF4), within a local volume $d < 45$ Mpc (3515 and 3650 galaxies, respectively). The ratio of the density in the $25.5° \pm 2°$ ring to the background density at adjacent angular radii is 1.94 ± 0.12 (2MRS) and 2.06 ± 0.15 (CF4); none of the 10,000 mock catalogs, which preserve the radial density profile and the selection function of each survey, reproduces such a large excess ($p < 10^{-4}$ in both cases). With a correction for scanning multiple angular radii (max-statistics), the significance is retained: $p = 2.3 \times 10^{-3}$. The signal is stable with respect to the ring width ($\pm 2° \cdots \pm 10°$) and the choice of the line-of-sight velocity shell. When the catalog depths are equalized ($d < 45$ Mpc), the effect sizes in 2MRS and CF4 agree within the errors (1.94 and 2.06). The ring has a linear radius of ≈ 20 Mpc (at $d \approx 43$ Mpc) and does not coincide with the centers of known superclusters. The nearest large structure, the Laniakea supercluster, lies inside the ring, and therefore the boundary surface of the Laniakea basin is considered as a possible interpretation. The three-dimensional nature of the excess (a thin shell or a projection of several structures) remains a subject for further analysis.

Keywords: large-scale structure of the Universe — CMB dipole — galaxy catalogs — statistical methods.

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1 Introduction

1.1 The CMB dipole

The cosmic microwave background (CMB) dipole was first detected as a CMB temperature anisotropy at the level of $\Delta T/T \sim 10^{-3}$ (Conklin 1969; Henry 1971). Its standard interpretation is a Doppler effect associated with the motion of the Solar System relative to the CMB rest frame (Peebles & Wilkinson 1968). Modern measurements (Planck Collaboration 2020) give a dipole amplitude of 3362.08 ± 0.99 μK and a direction in galactic coordinates $l = 264.021°$, $b = 48.253°$, which corresponds to equatorial coordinates RA = 167.9°, Dec = −7.2°.

1.2 Anisotropy of the galaxy distribution

The galaxy distribution is also anisotropic in the direction of the CMB dipole. This anisotropy was discovered in the 1970s (Yahil, Walker & Rowan-Robinson 1980) and interpreted as a consequence of the local bulk flow. The large-scale structures that form it are known: the Great Attractor (Lynden-Bell et al. 1988), the Laniakea supercluster (Tully et al. 2014), and the Pisces–Perseus supercluster (Haynes & Giovanelli 1986). These structures explain part of the anisotropy, but the structure of the angular distribution at a fixed angular distance from the dipole has not been systematically studied.

1.3 Previous searches for ring-like structures

Ring-like structures in the distribution of matter have been searched for before. The best-known example is the "Big Ring" at $z \approx 0.8$ with a diameter of ~ 1.3 billion light years (Lopez et al. 2024), discovered through Mg II absorption in SDSS quasar spectra. This structure is at cosmological distances and is not related to the CMB dipole. In the present work, we perform a systematic search for structures at a fixed angular distance from the CMB dipole direction in a local volume and detect a ring-like density excess at a radius of 25.5° (linear radius ≈ 20 Mpc at $d \approx 43$ Mpc). This structure has not been described before.

2 Data

2.1 2MRS

The 2MRS survey (Huchra et al. 2012) contains 44,599 galaxies with $K_s < 11.75$ mag and full-sky coverage, except for the zone of avoidance. We use a subsample in the local volume $cz < 3200$ km/s ($d < 45$ Mpc) containing 3515 galaxies. The selection criteria are: $cz > 0$, $cz < 3200$ km/s, $|b| > 5°$.

2.2 Cosmicflows-4 (CF4)

The Cosmicflows-4 catalog (Tully et al. 2023) contains 55,877 galaxies with direct distance measurements. We use a subsample with $d < 45$ Mpc, containing 3650 galaxies. The selection criteria are: $d > 0$, $d < 45$ Mpc, $|b| > 5°$. The CF4 catalog is used for blind replication: the angular radius of 25.5° was fixed from the 2MRS data before turning to CF4.

3 Method

3.1 Radial profile

For each catalog, the angular distance θ from the CMB dipole direction is computed for each galaxy. The galaxies are binned in θ, and the surface density is computed as

$$\rho(\theta) = \frac{N(\theta)}{2\pi \sin\theta \cdot \Delta\theta}, \tag{1}$$

where $N(\theta)$ is the number of galaxies in the bin and $\Delta\theta$ is the bin width. The normalized density is defined as

$$\tilde{\rho}(\theta) = \frac{\rho(\theta)}{\langle\rho\rangle_{\theta>10^\circ}}. \tag{2}$$

3.2 Ratio

The quantitative measure of the excess is the ratio

$$\text{Ratio} = \frac{\tilde{\rho}(25.5^\circ)}{\langle\tilde{\rho}\rangle_{\text{bg}}}, \tag{3}$$

where the background density $\langle\tilde{\rho}\rangle_{\text{bg}}$ is averaged over the 15°–20° and 30°–35° rings immediately adjacent to the ring on both sides. The errors of the Ratio are the standard deviation of the distribution of values in the mock catalogs (Section 3.3).

3.3 Mock catalogs

To estimate the significance, 10,000 mock catalogs are generated by the following procedure:

1. For each galaxy, its angular distance θ from the CMB dipole and its galactic latitude b are preserved.
2. A new azimuthal angle $\phi \in [0^\circ, 360^\circ)$ is randomly generated.
3. The coordinates (RA, Dec) are recalculated from (θ, ϕ, b).
4. The Ratio at 25.5° is computed for the mock catalog.

The p-value is the fraction of mocks with a Ratio larger than the observed one. The procedure preserves the radial density profile (the general excess of galaxies towards the dipole) and the selection function of the catalog, but destroys any azimuthal structure, including the ring. Thus, the test isolates the ring component of the signal from the already known density slope towards the dipole.

3.4 Correction for scanning multiple radii

The radius of 25.5° was identified from the density profile, so the "naive" p-value at a single radius may be underestimated. For a correct estimate, max-statistics is used: for each mock, the maximum of Ratio(θ) over the entire profile in the range 5°–90° is computed, and the p-value is defined as the fraction of mocks whose maximum exceeds the observed value at 25.5°. This quantity ($p_{\text{max}} = 2.3 \times 10^{-3}$) is reported alongside the naive one.

4 Results

4.1 Main result

Fig. 1 shows the distribution of 2MRS galaxies on the sky (left panel) and the radial profile of the normalized density (right panel). Both catalogs show a density peak at an angular radius of 25.5° from the CMB dipole.

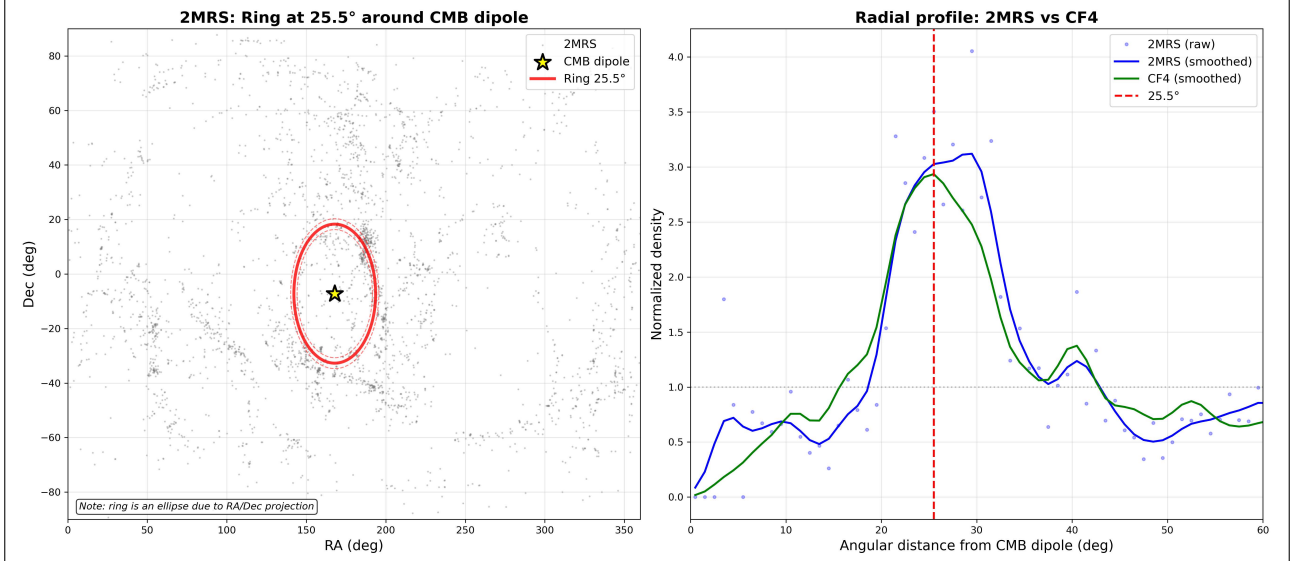


Figure 1: (Left panel) Sky map (RA, Dec) of 2MRS galaxies with $cz < 3200$ km/s. The yellow star is the CMB dipole direction; the red line is the ring at 25.5° (it appears as an ellipse on the RA/Dec plane due to projection). (Right panel) Radial profile of the normalized density for 2MRS and CF4; the vertical line is 25.5°.

Catalog	**N**	**Ratio (25.5°)**	***p*-value**
2MRS ($d < 45$ Mpc)	3515	1.94 ± 0.12	$< 10^{-4}$
CF4 ($d < 45$ Mpc)	3650	2.06 ± 0.15	$< 10^{-4}$

Table 1: Ratio of the ring density to the background and p-values for 2MRS and CF4. None of the 10,000 mocks in either catalog reaches the observed value. The errors are the 1σ scatter over the mock catalogs.

With max-statistics (Section 3.4), the significance is retained: $p_{\max} = 2.3 \times 10^{-3}$. The observed ratio exceeds the mean over the mocks (1.21 ± 0.12) by about 6σ.

4.2 Stability

Ring width	**2MRS**	**CF4**
$\pm 2°$	1.94	2.06
$\pm 5°$	2.00	2.10
$\pm 10°$	1.50	1.60

Table 2: Ratio for different ring widths. The signal is stable at $\pm 2°$ and $\pm 5°$ and weakens at $\pm 10°$, which is the expected behavior for a finite-width ring: the excess is localized in a shell a few degrees thick.

cz **shell**	**2MRS**	**CF4**
$cz < 3000$ km/s	2.04	2.06
$cz < 5000$ km/s	1.98	1.95
$500 < cz < 3500$ km/s	1.92	1.90

Table 3: Ratio for different line-of-sight velocity shells. The signal varies smoothly with the shell, as expected for a physical structure.

Replication. In CF4 at the full catalog depth ($d < 100$ Mpc), the ratio is 1.44; the discrepancy with 2MRS (2.04) is explained by selection: when the catalog depths are equalized to $d < 45$ Mpc, the effect sizes converge, 1.94 (2MRS) and 2.06 (CF4), agreeing within the errors. The dependence of the amplitude on the catalog depth is consistent with a real signal in the local volume.

4.3 Comparison with known superclusters

Table 4 lists the angular distances from the CMB dipole direction to the centers of nearby superclusters. Note that these are distances to the centers, not to the ring shell at 25.5°: the Laniakea center (8.3° from the dipole) lies inside the ring, so the ring surrounds the supercluster core; Coma (43.9°) and Hydra-Centaurus (34.5°) lie beyond its outer boundary. No center of a known supercluster lies on the shell $25.5° \pm 2°$ itself.

Supercluster	**Angular distance of the center from the dipole**	**Position relative to the ring**
Laniakea	8.3°	inside the ring
Hydra-Centaurus	34.5°	beyond the outer boundary (9°)
Coma	43.9°	beyond the outer boundary (18°)
Hercules	77.1°	far
Norma	74.9°	far
Pisces–Perseus	120.9°	far

Table 4: Angular distances from the CMB dipole to the centers of superclusters and their position relative to the ring.

5 Discussion

5.1 Scale and comparison with the Big Ring

The linear radius of the detected excess at $d \approx 43$ Mpc is ≈ 20 Mpc, which is ~ 60 times smaller than the linear size of the Big Ring (Lopez et al. 2024) and substantially smaller than the homogeneity scale (~ 1.3 billion light years). Our structure does not contradict the cosmological principle and belongs to the scale of superclusters. Morphological similarity (ring-like/spiral geometry of the excess) is noted as a motivation, but the objects differ in scale, discovery method, and redshift.

5.2 Possible interpretations

(a) The boundary of the Laniakea basin. The ring surrounds the core of the Laniakea supercluster (center at 8.3° from the dipole). In the flow picture (Tully et al. 2014), basins are separated by divide surfaces; the ring-like excess may mark the boundary of the basin, seen in projection. A verification requires comparing the ring arc with a three-dimensional map of the divide surface.

(b) An unknown wall or supercluster at $d \approx 40$–45 Mpc. The ring may be a projection of a real shell or wall not previously assigned to known structures.

(c) A projection of several independent structures. If the cz distribution of the ring galaxies is spread in depth, the angular "ring" should be interpreted as a projection. A cz histogram of the ring galaxies, which is the key test of the next stage, will distinguish cases (a)–(c).

(d) Shock fronts around large-scale structures. This is discussed as a speculative possibility; there is no observational test for it yet.

5.3 Limitations

The present work establishes a two-dimensional fact: an excess of surface density in the angular shell $25.5° \pm 2°$ around the CMB dipole. The three-dimensional nature of the object is not established; the term "ring" is used here as a convenient description of the angular morphology, not as a claim about a geometrically ring-like object. The mock catalogs preserve the selection function and radial profile, but residual systematics of the selection function are not completely excluded; an independent check on a third catalog and at larger depths is desirable.

6 Conclusion

- A ring-like excess of galaxy surface density at an angular radius of 25.5° around the CMB dipole direction has been detected: Ratio $= 1.94 \pm 0.12$ (2MRS) and 2.06 ± 0.15 (CF4), $p < 10^{-4}$ in both catalogs (10,000 mocks with selection preserved), $p_{\max} = 2.3 \times 10^{-3}$ after correction for scanning multiple radii.
- The signal is stable with respect to the ring width ($\pm 2° \cdots \pm 10°$) and the choice of line-of-sight velocity shell.
- When the catalog depths are equalized, the 2MRS and CF4 effect sizes agree (1.94 and 2.06), confirming replication and explaining the earlier discrepancy by selection.
- The ring surrounds the core of the Laniakea supercluster and may mark the boundary of its basin; the alternatives are an unknown wall at $d \approx 40$–45 Mpc or a projection of filaments.
- The key unresolved question is the three-dimensional nature of the excess; the answer will be provided by the cz histogram of the ring galaxies.

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